

TopoDistill: Topology-Aware Crack and Spalling Segmentation in Concrete Dams via Dynamic Snake Convolution and Dual-Teacher Distillation

Author A

Author B

Author C

Abstract

Automated semantic segmentation of cracks and spalling on concrete dam surfaces is essential for structural health monitoring, yet existing methods optimise pixel-level objectives without preserving the structural continuity of detected defects. We propose **TopoDistill**, a topology-aware segmentation framework incorporating two complementary enhancements to a SegFormer-B2 backbone: (1) a *Dynamic Snake Convolution* (DSConv) branch that captures elongated crack geometry through deformable sampling, adding only 4M parameters; and (2) *Class-Conditional Dual-Teacher Knowledge Distillation* (DTKD) that transfers complementary knowledge from a task-specific teacher and a foundation-model teacher with per-class weighting and pixel-level confidence modulation, at zero additional inference cost. On the **DamSegment** dataset (1,500 images, 3 classes), TopoDistill+DTKD achieves 72.3% foreground mIoU, the highest among all compared methods. Ablation experiments confirm complementary gains: DSConv drives crack connectivity ($\text{ConnR}_{\text{fg}} +3.4$), while DTKD drives pixel-level accuracy ($\text{mIoU}_{\text{fg}} +1.9$) and overall connectivity ($\text{ConnR}_{\text{fg}} +2.6$), also reducing cross-seed variance by $\sim 3\times$. We additionally investigate Skeleton Recall Loss (SRL), finding an asymmetric effect: SRL benefits teacher training but degrades the student when combined with DTKD ($-1.1 \text{ mIoU}_{\text{fg}}$), revealing that optimal training recipes differ between teachers and students. An exploration of alternative training strategies reveals a persistent crack–spalling tradeoff at shared class boundaries, and zero-shot cross-dataset evaluation on S2DS confirms that DSConv’s geometric inductive bias substantially improves cross-domain robustness (18.7% vs. 0.1% mIoU_{fg} for SegFormer-B2).

Dynamic Snake Convolution improves crack connectivity ($\text{ConnR}_{\text{fg}} +3.4$) with only 4M additional parameters.

Dual-teacher distillation boosts mIoU_{fg} by +1.9 at zero inference cost and reduces cross-seed variance by $\sim 3\times$; confidence-aware KD further improves crack BF1 by +0.8.

SRL benefits teacher training but harms student under distillation, revealing asymmetric training dynamics.

DSConv’s geometric bias enables 18.7% cross-dataset mIoU_{fg} vs. 0.1% for vanilla SegFormer.

Loss exploration reveals a persistent crack–spalling tradeoff at shared class boundaries.

Crack segmentation ,Concrete dam inspection ,Dynamic snake convolution ,Knowledge distillation ,Confidence-aware distillation ,Structural health monitoring

Introduction

Concrete dams are safety-critical infrastructure whose structural integrity must be continuously monitored throughout multi-decade service lives. Surface defects—predominantly cracks and spalling—serve as early indicators of internal deterioration, yet their manual inspection remains labour-intensive, subjective, and logistically constrained by

the scale and accessibility of dam faces . Automated semantic segmentation of these defects from high-resolution imagery therefore promises substantial gains in inspection efficiency and repeatability.

Recent deep-learning methods have advanced pixel-level defect segmentation considerably. Encoder-decoder architectures such as U-Net and hierarchical transformers like SegFormer achieve competitive Intersection-over-Union (IoU) on benchmark crack datasets . However, IoU and related region-overlap metrics are insensitive to topological errors: a prediction that fragments a continuous crack into disconnected segments may still score favourably on IoU while misrepresenting the structural extent of the defect . For structural safety assessment, the *connectivity* of a predicted crack is as important as its pixel coverage—a fragmented prediction can lead an engineer to underestimate propagation length and thus misjudge residual capacity.

Topology-preserving losses such as cIDice and persistent-homology penalties partially address this gap, yet they either incur prohibitive GPU memory for high-resolution dam images or operate on binary masks without multi-class support. The recently proposed Skeleton Recall Loss (SRL) overcomes both limitations by computing recall on CPU-extracted ground-truth skeletons with over 90% reduction in computational overhead. Concurrently, Dynamic Snake Convolution (DSConv) reshapes kernel sampling grids to follow elongated, tortuous paths—a morphological prior that aligns naturally with the geometry of cracks.

Meanwhile, large vision foundation models such as SAM and DINOv2 offer powerful pretrained representations, yet our experiments show that parameter-efficient fine-tuning via LoRA on small industrial datasets underperforms fully fine-tuned task-specific architectures. A natural question arises: *can the complementary strengths of task-specific and foundation models be combined?* We address this through knowledge distillation, where a student model learns from both a task-specific teacher and a foundation-model teacher with class-conditional weighting.

In this paper we propose **TopoDistill**, a framework that augments SegFormer-B2 with a parallel DSConv branch and class-conditional dual-teacher knowledge distillation (DTKD). We additionally investigate Skeleton Recall Loss (SRL) and uncover an asymmetric role in distillation: SRL enriches teacher training but harms the student when combined with DTKD. We evaluate on the semantic segmentation subset of the DamSegment dataset , comprising 1,500 annotated dam surface images. Our contributions are:

1. A topology-aware segmentation framework with two complementary enhancements: a parallel Dynamic Snake Convolution branch that drives crack connectivity ($\text{ConnR}_{\text{fg}} + 3.4$) through geometry-aware sampling, and a dual-teacher knowledge distillation scheme (DTKD) that drives pixel-level accuracy ($\text{mIoU}_{\text{fg}} + 1.9$ over DSConv alone) by transferring complementary knowledge from a task-specific teacher and a foundation-model teacher. DTKD incorporates confidence-aware distillation that modulates the KD loss at pixel level based on teacher agreement, improving boundary quality ($\text{BF1}_{\text{cr}} + 0.8$, $\text{cIDice}_{\text{cr}} + 0.6$) at no additional inference cost. The framework adds only 4M parameters with no inference-time overhead from distillation.
2. An empirical finding that Skeleton Recall Loss (SRL) exhibits an *asymmetric* effect in knowledge distillation: SRL benefits *teacher* training (producing richer representations that improve student performance by $+1.2 \text{ mIoU}_{\text{fg}}$) but harms *student* training when combined with DTKD ($-1.1 \text{ mIoU}_{\text{fg}}$), revealing that optimal training recipes differ between teachers and distillation students.
3. A comprehensive evaluation including comparison with six baselines (including SAM-LoRA and DINOv2-LoRA), per-tier difficulty analysis, statistical significance testing, zero-shot cross-dataset transfer on S2DS, and loss function/training strategy analysis revealing a persistent crack-spalling tradeoff at shared class boundaries, demonstrating that task-specific design with knowledge distillation outperforms direct foundation-model fine-tuning on small-scale industrial datasets and that DSConv’s geometric inductive bias substantially improves cross-domain robustness.

Related Work

Concrete defect segmentation

Early CNN-based crack detectors such as CrackNet treated defect identification as patch classification. Fully convolutional designs subsequently enabled dense pixel-wise prediction: DeepCrack introduced deeply-supervised side outputs, while exploited hierarchical features with multi-scale fusion.

Transformer-based architectures have demonstrated improved global context modelling. CrackFormer embeds self-attention to capture long-range crack continuity, and SegFormer provides a lightweight hierarchical encoder with an all-MLP decoder. Universal architectures such as Mask2Former further push segmentation accuracy on general benchmarks but have seen limited adoption for infrastructure defects due to annotation scarcity.

Applied to dam inspection specifically, performed multi-class concrete defect segmentation using DeepLabV3+, while proposed a lightweight CNN-Transformer hybrid for dam crack patrol inspection. surveyed visual perception methods for intelligent dam inspection, identifying limited dataset scale and poor topology preservation as key open challenges. More recently, CCT Net combined CNN and Transformer branches for dam surface crack segmentation, achieving 88.7% mIoU on a custom single-class dataset, though it does not address multi-class segmentation or topology preservation.

Topology-aware segmentation

introduced a differentiable topological loss based on persistent homology. extended persistent-homology supervision to cardiac imaging. However, persistent-homology computation scales poorly to high-resolution images.

For tubular structures, proposed cIDice, a differentiable soft-skeleton metric, but it requires GPU-based skeletonisation with substantial memory overhead. The Skeleton Recall Loss pre-computes ground-truth skeletons on CPU and formulates a recall objective, achieving superior connectivity with over 90% memory reduction relative to cIDice. Recent work has critically examined topology-aware losses: question whether SRL's benefits generalise beyond the original evaluation setting, and provide a faithful implementation of the Crack Topology Score with extensions for handling common prediction artifacts.

Our work evaluates SRL in the multi-class dam defect setting, revealing an asymmetric effect: SRL enriches teacher representations for distillation but harms student training when combined with soft-label objectives—consistent with the growing evidence that topology-aware losses require careful integration.

Deformable and dynamic convolutions

Deformable Convolutional Networks (DCN) introduced learnable offsets for adaptive sampling. Dynamic Snake Convolution (DSConv) constrains offsets along a one-dimensional path, forcing snake-like sampling that traces elongated structures. DSConv has since been adopted for various crack detection tasks: applied it to metro tunnel lining crack detection, and proposed a dual-branch architecture combining enhanced DSConv with SegFormer for road crack segmentation, achieving competitive IoU on Crack3238 and FIND datasets.

Our work differs from in three aspects: (1) we address multi-class segmentation (crack and spalling) rather than binary crack detection; (2) we introduce class-conditional dual-teacher knowledge distillation (DTKD) with confidence-aware pixel-level modulation, which is absent from their architecture-only approach; and (3) we target dam-specific inspection with topology-aware evaluation (ConnR, cIDice) and cross-dataset transfer analysis.

Knowledge distillation for segmentation

Knowledge distillation (KD) transfers knowledge from a teacher model to a smaller student

by matching softened output distributions. In semantic segmentation, structured KD approaches distil spatial relationships rather than just per-pixel logits. Multi-teacher distillation combines knowledge from multiple specialised teachers, but typically uses uniform teacher weighting across all classes. proposed confidence-aware multi-teacher KD (CA-MKD) with sample-wise teacher reliability weighting, though at the sample level rather than per-class or per-pixel. Recently, demonstrated task-specific KD from vision foundation models for medical segmentation, showing that distilling SAM’s knowledge into lightweight students outperforms direct fine-tuning—a finding consistent with our results in the infrastructure domain.

Our class-conditional dual-teacher strategy differs by assigning per-class teacher weights based on each teacher’s domain-specific strengths, combined with pixel-level confidence modulation based on teacher disagreement. This contrasts with CA-MKD’s sample-level weighting and with ’s single-teacher approach.

Foundation models for segmentation

The Segment Anything Model (SAM) and DINOv2 represent powerful vision foundation models. Parameter-efficient fine-tuning via LoRA enables adaptation with minimal trainable parameters ($\sim 5\%$). While promising on natural images, their effectiveness on small-scale industrial datasets remains underexplored. applied SAM-based instance segmentation for structural damage detection, demonstrating SAM’s potential for infrastructure inspection but also noting the need for domain-specific adaptation. We show that rather than directly deploying foundation models, using them as teachers in knowledge distillation yields better results—a strategy that preserves inference efficiency while inheriting the foundation model’s structural knowledge.

Method

Overview

Fig. [fig:architecture] illustrates the TopoDistill framework. Our approach enhances a SegFormer-B2 backbone at two complementary levels: (1) *feature extraction* via a parallel DSConv branch for geometry-aware crack sampling, and (2) *knowledge transfer* via class-conditional dual-teacher distillation (DTKD) to combine the strengths of task-specific and foundation models.

image

SegFormer-B2 backbone

We adopt SegFormer-B2 as our backbone, providing a hierarchical Mix Transformer (MiT) encoder with four stages at resolutions $H/4$, $H/8$, $H/16$, and $H/32$ with channel dimensions 64, 128, 320, and 512. The all-MLP decoder fuses multi-scale features via linear projection and upsampling. We initialise from weights pretrained on ADE20K at 512×512 .

Dynamic Snake Convolution branch

Standard convolutions sample on fixed rectangular grids—poorly matched to thin, curvilinear cracks. Inspired by , we introduce a parallel *CrackSnakeBranch* that performs deformable sampling along crack geometry.

Architecture.

The branch consumes encoder stages 1 and 2 (at $H/4$ and $H/8$ resolution). Two parallel DSConv paths deform sampling along the x -axis and y -axis respectively, each with kernel size $K = 9$ and hidden dimension $d = 64$. The outputs are projected to a single-channel

crack enhancement map $\mathbf{E}_{\text{crack}} \in \mathbb{R}^{H \times W}$.

Additive enhancement.

The map is added *only* to the crack channel logits:

$$\hat{\mathbf{y}}_{\text{crack}} = \mathbf{z}_{\text{crack}} + \mathbf{E}_{\text{crack}} \quad \text{\label{eq:dscnv}}$$

where $\mathbf{z}_{\text{crack}}$ denotes the decoder’s crack logits. Spalling and background logits remain unmodified.

Zero-initialisation.

The output projection is initialised to zero, so the branch starts as a no-op and gradually learns crack-specific deformations during training.

Class-conditional dual-teacher knowledge distillation

A key observation from our experiments is that different model architectures excel at different defect types: the task-specific TopoDistill (Teacher 1) achieves higher spalling IoU, while the foundation-model SAM-LoRA (Teacher 2) achieves higher crack connectivity (ConnR). Standard knowledge distillation cannot exploit this complementarity because it applies uniform teacher weighting across all classes.

We propose *class-conditional dual-teacher knowledge distillation* (DTKD) that constructs a per-class ensemble of teacher predictions:

Teacher models.

We use two pretrained teachers, both trained with Skeleton Recall Loss (SRL) to enrich their learned representations (the rationale for this choice is validated in Section 4.6):

- **Teacher 1** (T_1): TopoDistill (task-specific, strong on spalling)
- **Teacher 2** (T_2): SAM-LoRA (foundation model, strong on crack connectivity)

We select SAM-LoRA over DINOv2-LoRA as Teacher 2 because SAM-LoRA achieves substantially higher ConnR_{fg} (81.2% vs. 76.5%) and BF1_{fg} (68.1% vs. 62.4%), providing the complementary connectivity knowledge that the task-specific teacher lacks (Table [tab:main]). DINOv2-LoRA’s advantage is limited to IoU_{sp} , which Teacher 1 already covers.

Class-conditional ensemble.

For each class c , the ensemble teacher probability is computed as a weighted combination:

$$\tilde{p}_c = w_{1,c} \cdot p_{1,c} + w_{2,c} \cdot p_{2,c}$$

where $p_{i,c} = \text{softmax}(\mathbf{z}_i/\tau)_c$ is the softened probability from teacher i at temperature τ , and $w_{1,c} + w_{2,c} = 1$ for each class.

Class-conditional weights were determined based on per-class performance analysis of the two teachers on the validation set and fixed before test-set evaluation. Since Teacher 2 (SAM-LoRA) achieves higher ConnR_{cr} while Teacher 1 achieves higher IoU_{sp} , we assign higher Teacher 2 weight for crack and higher Teacher 1 weight for spalling:

- **Background:** $w_{1,\text{bg}} = w_{2,\text{bg}} = 0.5$ (equal, neither teacher specialises)
- **Crack:** $w_{2,\text{cr}} = 0.6$ (SAM-LoRA has better crack connectivity)
- **Spalling:** $w_{2,\text{sp}} = 0.3$ (TopoDistill has better spalling coverage)

The equal-weight baseline (Table 5) shows that this assignment provides modest mIoU gains (+0.1) but notably improves crack centreline preservation (cDice_{fg} +0.6). Manual weight selection is a limitation; adaptive class-conditional weighting is discussed in Section 5.

Distillation loss.

The ensemble probabilities are renormalised to sum to one and the student is trained with:

$$\mathcal{L}_{\text{KD}} = \tau^2 \cdot \text{KL}(\tilde{\mathbf{p}} \parallel \text{softmax}(\mathbf{z}_s / \tau)) \quad \text{eq:kd_loss}$$

where $\tilde{\mathbf{p}}$ denotes the renormalised class-conditional ensemble probability and $\mathbf{z}_s$ the student logits.

Confidence-aware distillation.

When two teachers with different architectures and training recipes are combined, their predictions inevitably disagree at certain pixels—particularly at ambiguous crack–background boundaries and thin structures. At such pixels the ensemble soft label is unreliable, and forcing the student to match it may introduce noise. We quantify per-pixel teacher agreement using the KL divergence between the two softened teacher distributions:

$$d(i) = \text{KL}(\text{softmax}(\mathbf{z}_{T_1}^{(i)}/\tau) \parallel \text{softmax}(\mathbf{z}_{T_2}^{(i)}/\tau))$$

where i indexes spatial positions. An agreement map is derived as $a(i) = 1 - d(i)/\max_j d(j)$, ranging from 1 (full agreement) to 0 (maximum disagreement). The confidence-aware distillation loss modulates the per-pixel KD contribution:

$$\mathcal{L}_{\text{KD}}^{\text{conf}} = \tau^2 \cdot \frac{1}{N} \sum_i a(i) \cdot \text{KL}(\tilde{\mathbf{p}}^{(i)} \parallel \text{softmax}(\mathbf{z}_s^{(i)} / \tau)) \quad \text{eq:conf_kd}$$

This ensures the student learns strongly from the ensemble where both teachers concur, but falls back on the ground-truth supervised loss at ambiguous pixels. The agreement map is computed within the existing forward pass (no additional inference cost). Fig. 1 visualises the disagreement map on representative Hard-tier samples: high-disagreement pixels (bright regions) concentrate at crack–background boundaries and thin branching points, precisely where the ensemble soft label is least reliable. We empirically validate that teacher disagreement correlates with sample difficulty: the mean per-sample KL divergence between teachers is 43% higher on Hard-tier images than on Easy-tier images (Section 4.6).

□ Teacher disagreement visualisation on Hard-tier samples. Columns: input, ground truth, Teacher 1 (TopoDistill) prediction, Teacher 2 (SAM-LoRA) prediction, pixel-level KL divergence heatmap (bright = high disagreement), student (DTKD) prediction. Disagreement concentrates at crack boundaries and thin structures where the ensemble soft label is unreliable. Red: crack, green: spalling.

Total loss.

The final training objective balances the segmentation loss and distillation loss:

$$\mathcal{L} = (1-\alpha)[\lambda_1 \mathcal{L}_{\text{CE}} + \lambda_2 \mathcal{L}_{\text{Dice}}] + \alpha \mathcal{L}_{\text{KD}}^{\text{conf}}$$

with $\alpha = 0.5$, $\tau = 4$, and $\lambda_1 = \lambda_2 = 0.5$. Note that SRL is used only during *teacher* training; the student does not use SRL, as we find it conflicts with the distillation objective (Section 4.6).

Test-time augmentation

At inference, we optionally employ multi-scale TTA: each image is evaluated at three scales (0.75×, 1.0×, 1.25×) with horizontal and vertical flips (9 views). Logits are averaged

before argmax.

Experiments

Dataset

We evaluate on the **semantic segmentation subset** of the DamSegment dataset, which comprises 3,500 image patches captured during an on-site visit of a concrete dam, supporting classification, detection, and segmentation tasks. The segmentation subset contains 1,500 pixel-level annotated images with three classes (background, crack, spalling) at 512×512 resolution, organised into three difficulty tiers (Easy, Medium, Hard; 500 each) based on crack detection difficulty, enabling curriculum learning. We split the data into train/val/test sets of 1,200/150/150 images, stratified by tier. Splitting is at the patch level; patches from adjacent regions may share visual characteristics (see Section 5). Fig. 2 shows representative samples from each tier with ground-truth overlays; crack pixels range from 0.3% to 7.9% and spalling from 1.0% to 11.7%, illustrating the extreme class imbalance.

□
DamSegment dataset samples by difficulty tier. Each pair shows the raw image (left) and ground-truth overlay (right). Red: crack, green: spalling. Crack and spalling pixel ratios are annotated above each overlay. Easy samples contain wide, well-separated cracks; Hard samples feature thin, branching cracks with co-occurring spalling.

Implementation details

All models are trained on a single NVIDIA A40 GPU using PyTorch. TopoDistill uses SegFormer-B2 pretrained on ADE20K-512, trained for 100 epochs with AdamW (lr 6×10^{-5} , weight decay 10^{-4}), 5-epoch warmup, batch size 4. The DSConv branch uses $d = 64$, $K = 9$. Data augmentation consists of random horizontal and vertical flips ($p = 0.5$ each); images are resized to 512×512 and normalised with ImageNet statistics.

For DTKD, Teacher 1 (TopoDistill+SRL) and Teacher 2 (SAM-LoRA+SRL) are trained independently first with SRL ($\lambda_3 = 0.05$, activated at epoch 60), then frozen during student training. The student has identical architecture to Teacher 1 but is trained *without* SRL (CE + Dice + KD only). Distillation uses $\alpha = 0.5$, $\tau = 4$.

For foundation model baselines, SAM-LoRA uses ViT-B with LoRA (rank 16, $\alpha=32$) and an FPN decoder. DINOv2-LoRA adapts ViT-B/14 similarly with input size 518×518 (patch-size multiple). Both train for 50 epochs with batch size 2.

Evaluation metrics

We report complementary metrics capturing different aspects of segmentation quality:

- **mIoU_{fg}**: Mean IoU of crack and spalling classes.
- **IoU_{cr}, IoU_{sp}**: Per-class IoU.
- **BF1**: Boundary F1 score (2-pixel tolerance) .
- **cdDice**: Centreline Dice via skeletonisation .
- **ConnR**: Connectivity Ratio—fraction of GT connected components with $\geq 50\%$ overlap in prediction.

Computational cost

Table 1 summarises the model complexity and inference speed measured on a single NVIDIA A40 GPU at each model’s native input resolution. TopoDistill adds 4.0M parameters (+14.5%) over SegFormer-B2 due to the DSConv branch, with a modest FPS reduction (124→96). Crucially, DTKD incurs *no additional inference cost*: the two teachers are only used during training, and the student has identical architecture to the base TopoDistill. Mask2Former is the heaviest model (68.7M, 39 FPS), while foundation model baselines (SAM-LoRA, DINOv2-LoRA) carry >90M parameters due to their ViT-B encoders, though only ~5–6M are trainable via LoRA.

Computational cost comparison. Params: total parameters. Train: trainable parameters (all are fully trainable unless noted). FPS: images/second on a single A40 GPU (batch size 1). †: LoRA fine-tuning. ^aSAM-LoRA requires prompt-based inference incompatible with our batch FPS benchmark; we omit this value rather than report a non-comparable measurement.

Model	Backbone	Params (M)	Train (M)	FPS
U-Net	ResNet-34	24.4	24.4	519
DeepLabV3+	ResNet-50	26.7	26.7	333
SegFormer-B2	MiT-B2	27.3	27.3	124
Mask2Former	Swin-S	68.7	68.7	39
SAM-LoRA [†]	ViT-B	93.7	5.0	— ^a
DINOv2-LoRA [†]	ViT-B/14	92.7	6.1	68
TopoDistill	MiT-B2	31.3	31.3	96
TopoDistill+DTKD	MiT-B2	31.3	31.3	96

Comparison with state-of-the-art methods

Table [tab:main] presents quantitative results on the DamSegment test set.

Method	Backbone	mIoU _{fg}	IoU _{cr}	IoU _{sp}	BF1 _{cr}	BF1 _{sp}	BF1 _{fg}	clDice
U-Net [†]	ResNet-34	59.6	51.3	67.9	75.4	53.8	64.6	75.3
DeepLabV3+	ResNet-50	63.4	53.3	73.5	69.0	49.5	59.2	78.2
Mask2Former	Swin-S	67.9	57.1	78.6	76.0	69.0	72.5	82.2
SegFormer-B2	MiT-B2	70.4	56.9	83.8	73.7	59.8	66.8	81.5
SAM-LoRA [†]	ViT-B	68.1	55.9	80.2	73.1	63.0	68.1	81.3
DINOv2-LoRA [†]	ViT-B	66.7	49.1	84.2	64.7	60.1	62.4	77.4
TopoDistill ^{SS}	MiT-B2	70.6	57.3	83.8	74.7	66.0	70.3	82.5
TopoDistill + DTKD	MiT-B2	<u>72.3</u>	<u>58.5</u>	<u>86.2</u>	<u>75.6</u>	65.6	70.6	<u>83.3</u>
TopoDistill + DTKD + TTA [*]	MiT-B2	72.7	59.3	86.0	76.3	<u>67.2</u>	<u>71.8</u>	83.7

TopoDistill with DTKD achieves the highest mIoU_{fg} (72.3% without TTA, 72.7% with TTA) and the highest IoU for both crack and spalling classes among all compared methods. Mask2Former achieves the highest BF1_{fg} (72.5%) due to its mask-attention mechanism, though with substantially lower mIoU_{fg} (67.9%). TopoDistill+DTKD+TTA surpasses Mask2Former on BF1_{cr} (76.3% vs. 76.0%). TopoDistill+DTKD achieves ConnR_{fg} 83.0%, surpassing SAM-LoRA (81.2%); this connectivity reflects complementary contributions from DSConv’s geometric inductive bias (+3.4 ConnR_{fg}) and DTKD’s improved spalling connectivity (+2.6 ConnR_{fg}). Compared to the SegFormer-B2 baseline, TopoDistill+DTKD improves mIoU_{fg} by +1.9, IoU_{cr} by +1.6, IoU_{sp} by +2.4, and ConnR_{fg} by +2.1.

Foundation model comparison.

Despite leveraging powerful pretrained representations, SAM-LoRA (68.1%) and DINOv2-LoRA (66.7%) substantially underperform both the base TopoDistill (70.6%) and TopoDistill+DTKD (72.3%). However, DTKD demonstrates that foundation model knowledge can be *indirectly* leveraged through distillation—a more effective use of foundation models on small-scale industrial datasets (see ablation in Section 4.6). Mask2Former achieves the highest BF1_{fg} (72.5%) and $\text{cIDice}_{\text{sp}}$ (91.1%), but with substantially lower mIoU_{fg} (67.9%).

Ablation study

Table [tab:ablation] presents the incremental contribution of each component.

Configuration	mIoU_{fg}	IoU_{cr}	IoU_{sp}	BF1_{cr}	BF1_{fg}	$\text{cIDice}_{\text{cr}}$	$\text{cIDice}_{\text{fg}}$	ConnR
(a) SegFormer-B2	70.3	56.9	83.7	74.0	68.6	81.8	82.7	81.0
(b) + DSConv	70.4	57.4	83.4	75.0	70.6	81.8	85.7	83.8
(c) + DTKD	72.3	58.5	86.2	75.6	70.6	83.3	86.4	<u>82.1</u>
(g) + conf-aware KD	<u>72.5</u>	<u>58.7</u>	86.3	76.4	<u>71.1</u>	83.9	<u>86.1</u>	81.4
(d) (c) + TTA	72.7	59.3	<u>86.0</u>	<u>76.3</u>	71.8	<u>83.7</u>	86.2	79.4
(e) (b) + SRL	70.6	57.3	83.8	74.7	70.3	82.5	85.7	82.2
(f) (e) + DTKD	71.2	58.0	84.5	75.4	71.4	83.2	86.2	82.3

(a)→(b): Adding DSConv.

The DSConv branch produces minimal mIoU change (+0.1) but substantially improves crack connectivity: ConnR_{cr} increases by **+2.8** (81.0→83.8) and $\text{cIDice}_{\text{fg}}$ by **+3.0** (82.7→85.7), along with improved boundary quality (BF1_{cr} +1.0, BF1_{fg} +2.0). The deformable snake-shaped sampling captures elongated crack geometry that fixed rectangular kernels miss, demonstrating that DSConv’s primary value lies in its geometric inductive bias for connectivity.

(b)→(c): Adding DTKD.

DTKD provides substantial pixel-level improvement: mIoU_{fg} +1.9, IoU_{cr} +1.1, IoU_{sp} +2.8, $\text{cIDice}_{\text{cr}}$ +1.5, and BF1_{cr} +0.6, while also improving overall connectivity: ConnR_{fg} +2.6 (80.4→83.0). ConnR_{cr} decreases modestly from 83.8 to 82.1 (−1.7), as DTKD’s soft-label supervision slightly smooths fine crack structures, but this is more than offset by substantially improved spalling connectivity. The class-conditional weighting ensures that the student receives the most informative teacher guidance for each defect type (Section 3.4).

(c)→(g): Confidence-aware KD.

Replacing the standard KD loss with confidence-aware distillation (Eq. [eq:conf_kd]) improves boundary quality: BF1_{cr} +0.8 (75.6→76.4) and $\text{cIDice}_{\text{cr}}$ +0.6 (83.3→83.9), with a small mIoU_{fg} gain (+0.2). This confirms that down-weighting the distillation signal at high-disagreement pixels reduces noise from unreliable ensemble labels. ConnR_{cr} decreases modestly (82.1→81.4), as the reduced KD weight at ambiguous boundary pixels allows the student to rely more on ground truth—which improves pixel precision but slightly reduces connectivity.

To isolate the contribution of confidence-aware KD from difficulty-guided sample reweighting (an alternative use of teacher disagreement), we conducted an ablation using disagreement-based per-sample loss reweighting *without* pixel-level confidence modulation. This variant degraded performance (mIoU_{fg} 72.1, BF1_{fg} 70.0), indicating that on small

datasets, per-sample reweighting introduces training instability, while pixel-level modulation of the distillation signal is the effective mechanism.

Analysis of teacher disagreement across difficulty tiers reveals that the mean per-sample KL divergence between teachers is 0.046 on Hard-tier images vs. 0.032 on Easy-tier images (+43%), confirming that disagreement is an informative proxy for sample difficulty. However, this correlation does not translate into effective sample-level curriculum learning—consistent with prior findings that data-restricting or reweighting curriculum strategies underperform on small datasets where every sample matters.

(c)→(d): Test-time augmentation.

TTA provides modest additional gains ($\text{mIoU}_{\text{fg}} +0.4$, $\text{IoU}_{\text{cr}} +0.8$) but reduces connectivity metrics ($\text{ConnR}_{\text{fg}} 83.0 \rightarrow 80.2$), suggesting that multi-scale/flip aggregation improves pixel coverage but may smooth fine structural details.

Effect of SRL (rows e–f).

Skeleton Recall Loss (SRL) penalises low predicted probability at ground-truth skeleton pixels $\square$, precomputed offline via morphological skeletonisation:

$$\mathcal{L}_{\text{SRL}} = 1 - \frac{1}{|\mathcal{S}|} \sum_{p \in \mathcal{S}} \hat{y}_{\text{crack}}(p) \quad \text{label{eq:srl}}$$

We apply SRL only to the crack class (activated at epoch 60, weight $\lambda_3 = 0.05$). Rows (e)–(f) show the effect of adding SRL to the pipeline. Adding SRL to DSConv (row e) provides no crack improvement ($\text{IoU}_{\text{cr}} -0.1$, $\text{BF1}_{\text{cr}} -0.3$) and damages connectivity: ConnR_{fg} drops from 80.4 to 77.8 (−2.6), driven by ConnR_{sp} falling from 77.0 to 73.4 as SRL’s crack-focused supervision fragments spalling components at ambiguous boundaries. Adding DTKD on top of SRL (row f) partially recovers connectivity (77.8→80.6) but the resulting model (f, $\text{mIoU}_{\text{fg}} 71.2\%$) underperforms the SRL-free pipeline (c, $\text{mIoU}_{\text{fg}} 72.3\%$) by **−1.1**. This demonstrates that SRL’s gradient interference with DTKD’s soft-label supervision is a net negative: the student learns better from teacher signals alone without the competing skeleton objective. Multi-seed experiments (Table 2) further confirm that the SRL path is unstable (mIoU_{fg} std 2.42 vs. 0.52 for DTKD alone).

Overall progression.

From baseline (a) to the full method with confidence-aware KD (g), the cumulative improvement is +2.2 mIoU_{fg} , +2.5 BF1_{fg} , +3.4 $\text{cIDice}_{\text{fg}}$, and +5.6 ConnR_{fg} , confirming that the three components contribute complementary gains.

Statistical significance.

To verify that DTKD’s gains are not due to random variation, we conduct paired Wilcoxon signed-rank tests on the test images, comparing DSConv-only (row b) vs. DSConv+DTKD (row c, final model). Six of ten per-class metrics reach significance at $\alpha = 0.05$: IoU_{cr} ($p = 0.008$), Dice_{cr} ($p = 0.007$), BF1_{cr} ($p = 0.046$), $\text{cIDice}_{\text{cr}}$ ($p < 0.001$), IoU_{sp} ($p = 0.016$), and Dice_{sp} ($p = 0.016$). The four non-significant metrics (BF1_{sp} , $\text{cIDice}_{\text{sp}}$, ConnR_{cr} , ConnR_{sp}) all involve small paired-sample sizes ($n = 31\text{--}37$) and show directionally consistent improvements, except ConnR_{cr} where DSConv-only is marginally better (−1.7, $p = 0.040$). Tier-stratified analysis reveals the significance concentrates on Hard images: IoU_{cr} ($p = 0.003$), Dice_{cr} ($p = 0.004$), BF1_{cr} ($p = 0.045$), and $\text{cIDice}_{\text{cr}}$ ($p < 0.001$) all reach significance, confirming that the foundation-model teacher’s global knowledge provides the strongest benefit on complex crack patterns.

Multi-seed training stability.

To assess robustness to random initialisation, we train three key configurations—SegFormer-B2 (baseline), TopoDistill+SRL, and TopoDistill+DTKD (row c, final model)—with three independent seeds (42, 123, 2024) and report mean $\pm$ standard deviation in Table 2.

Multi-seed stability analysis (mean $\pm$ std over 3 seeds).
TopoDistill+DTKD is the final model (row c, no SRL). All values in %.

Metric	SegFormer-B2	TopoDistill+SRL	TopoDistill+DTKD
mIoU _{fg}	69.2 $\pm$ 1.47	67.9 $\pm$ 2.42	71.8 $\pm$ 0.52
IoU _{cr}	55.5 $\pm$ 2.08	53.7 $\pm$ 3.89	58.5 $\pm$ 0.31
IoU _{sp}	82.9 $\pm$ 0.86	82.2 $\pm$ 0.95	85.1 $\pm$ 1.07
BFI _{fg}	67.1 $\pm$ 1.62	68.5 $\pm$ 2.84	71.1 $\pm$ 0.85
clDice _{fg}	81.8 $\pm$ 1.93	82.1 $\pm$ 3.44	86.3 $\pm$ 0.34
ConnR _{fg}	80.3 $\pm$ 2.13	75.7 $\pm$ 3.68	81.3 $\pm$ 1.43

DTKD achieves the highest mean performance on all six metrics while exhibiting the lowest variance on four of six. Notably, TopoDistill+SRL shows the highest variance (mIoU_{fg} std = 2.42), and its multi-seed mean (67.9%) is *lower* than the baseline SegFormer-B2 (69.2%), indicating that the single-seed improvement reported in Table [tab:ablation] is not robust. DTKD reduces sensitivity to initialisation by $\sim 3\times$ compared to the baseline (mIoU_{fg} std: 0.52 vs. 1.47), likely because the soft-label supervision from two pretrained teachers smooths the loss landscape.

Per-tier breakdown.

Table 3 reports per-image mean performance stratified by difficulty tier, comparing DSConv-only (row b) and DSConv+DTKD (row c, final model) to isolate DTKD’s contribution. DTKD improves mIoU_{fg} consistently across all tiers (+3.4 Easy, +2.3 Medium, +2.4 Hard), but the gains reach statistical significance only on Hard images (IoU_{cr} $p = 0.003$, clDice_{cr} $p < 0.001$), where complex crack patterns benefit most from the foundation-model teacher. The spalling improvements are largest on Easy (+6.2 IoU_{sp}, $p = 0.035$) where the SAM-LoRA teacher’s global context helps distinguish large spalling regions. ConnR_{fg} improves on Easy (+4.8) and Hard (+1.4) tiers, driven by ConnR_{sp} gains, while ConnR_{cr} shows a small decrease on Hard (-2.4 , $p = 0.018$)—a trade-off where DTKD slightly sacrifices crack connectivity for substantially better spalling connectivity and overall accuracy.

Per-tier performance breakdown (per-image means). Bold: better value per tier. Wilcoxon p -values reported in text. All values in %.

Tier	Model	mIoU _{fg}	IoU _{cr}	IoU _{sp}	BFI _{fg}	clDice _{fg}	ConnR _{fg}
Easy	DSConv (b)	62.2	59.8	64.6	66.8	86.7	84.2
	DSConv+DTKD (c)	65.6	60.4	70.8	68.9	88.8	89.0
Medium	DSConv (b)	56.2	59.7	52.7	77.4	87.3	74.8
	DSConv+DTKD (c)	58.5	60.8	56.2	78.4	90.5	74.2
Hard	DSConv (b)	61.9	52.8	71.0	72.6	84.1	81.7
	DSConv+DTKD (c)	64.2	53.8	74.6	74.1	85.5	83.1

Loss function and training strategy exploration

Error analysis.

To guide loss function design, we analyse the confusion matrix of the DTKD model (row c). For crack pixels, precision is 68.8% while recall is 79.7%, yielding a false-positive to false-negative ratio of 1.78:1. The majority of false positives occur at crack-background boundaries, where thin crack predictions leak into adjacent background pixels. This precision bottleneck motivates exploring loss functions that penalise false positives more heavily.

Strategy variants.

Table [tab:strategy] summarises six training strategies applied on top of the DTKD pipeline (row c), targeting crack precision improvement through loss modification or sampling adjustment. All variants use the same DSCFormer architecture with DTKD; they differ only in the segmentation loss or sampling strategy. Tversky loss generalises Dice by introducing asymmetric FP/FN weights ($\alpha > 0.5$ penalises false positives more heavily); WABL (Width-Aware Boundary Loss) applies an adaptive CE penalty at thin-crack boundary pixels where width falls below a threshold; Boundary Dice restricts the Dice computation to a dilated boundary region around ground-truth contours.

Strategy	mIoU _{fg}	IoU _{cr}	IoU _{sp}	BF1 _{cr}	BF1 _{sp}	BF1 _{fg}	clDice _{fg}	ConnR _{fg}
(c) DTKD (CE+Dice)	72.3	58.5	86.2	75.6	65.6	70.6	86.4	83.0
+ Tversky ($\alpha = 0.6$)	71.7	58.3	85.1	75.8	69.0	72.4	86.7	79.4
+ Tversky ($\alpha = 0.5$)	72.1	58.9	85.4	76.4	65.7	71.0	86.5	81.1
+ Boundary Dice	72.1	58.8	85.4	75.7	69.2	72.4	86.4	83.2
+ WABL	71.6	58.4	84.8	75.9	64.0	70.0	84.9	79.5
+ WABL + Tversky	71.0	58.6	83.4	76.0	63.7	69.9	84.0	79.0
Hard-tier FT + WABL	71.6	58.2	85.0	75.7	68.7	72.2	87.3	82.9

Crack-spalling tradeoff.

A consistent pattern emerges across all loss-based strategies: improving crack boundary precision degrades spalling metrics. Tversky loss with $\alpha = 0.6$ (stronger FP penalty) improves BF1_{sp} by +3.4 over the baseline, but reduces mIoU_{fg} by -0.6 and ConnR_{fg} by -3.6. WABL, which specifically targets thin-crack boundary false positives, causes the largest spalling regression (IoU_{sp} -1.4, BF1_{sp} -1.6). Combining WABL with Tversky amplifies the degradation (IoU_{sp} -2.8). This tradeoff arises because crack and spalling often co-occur at shared boundaries; loss terms that suppress crack false positives inadvertently penalise adjacent spalling predictions.

Hard-tier fine-tuning.

Sampling-based approaches partially mitigate this tradeoff. Fine-tuning from the DTKD checkpoint with 70% Hard-tier oversampling and WABL achieves BF1_{fg} 72.2 (comparable to the best loss-based variant) while maintaining better spalling coverage (IoU_{sp} 85.0 vs. 83.4-85.4 for loss variants) and the highest clDice_{fg} (87.3). However, no strategy surpasses the base DTKD model on mIoU_{fg} (72.3), indicating that the standard CE+Dice combination is already near-optimal for this dataset and that the crack precision bottleneck requires fundamentally different approaches (e.g., higher-resolution inputs or explicit boundary modelling in the architecture).

Analysis of DTKD teacher contributions

Table 4 compares individual teachers with the DTKD student.

Individual teacher performance vs. DTKD student on the test set. The student surpasses both teachers, confirming effective knowledge combination. All values in %.

Model	mIoU _{fg}	IoU _{cr}	IoU _{sp}	ConnR _{cr}	ConnR _{sp}
Teacher 1 (TopoDistill)	70.6	57.3	83.8	82.2	73.4
Teacher 2 (SAM-LoRA+SRL)	69.0	55.9	82.1	82.2	86.0
Student (DTKD)	72.3	58.5	86.2	82.1	<u>83.8</u>

The DTKD student surpasses both teachers on $mIoU_{fg}$ (+1.7 over Teacher 1, +3.3 over Teacher 2), IoU_{cr} and IoU_{sp} , confirming that class-conditional distillation effectively combines complementary strengths. Teacher 2 (SAM-LoRA+SRL) achieves the highest $ConnR_{sp}$ (86.0%), reflecting SAM’s strong global structure preservation. The student substantially inherits this advantage ($ConnR_{sp} = 83.8\%$, vs. Teacher 1’s 73.4%), while maintaining superior pixel-level accuracy. However, the student does not fully match Teacher 2’s $ConnR_{sp}$ (83.8% vs. 86.0%) and slightly trails both teachers on $ConnR_{cr}$ (82.1% vs. 82.2%), consistent with the $ConnR_{cr}$ trade-off observed in the ablation.

The student surpassing both teachers is consistent with multi-teacher distillation findings in the literature, where combining complementary knowledge sources provides regularisation benefits beyond what any single teacher offers. Unlike classical “born-again” networks where student and teacher share the same architecture, our student additionally benefits from the DSConv geometric inductive bias that neither teacher possesses.

Class-conditional weighting ablation.

Table 5 compares equal-weight (0.5/0.5) and class-conditional teacher weighting in the final SRL-free pipeline. The two configurations achieve nearly identical $mIoU_{fg}$ (72.3 vs. 72.2), but class-conditional weighting improves crack centreline preservation: $clDice_{fg}$ +0.6 (85.8→86.4), at the cost of slightly lower $BF1_{fg}$ (−0.9). The result suggests that class-conditional weighting primarily steers the distillation signal toward better crack structure preservation rather than improving overall pixel accuracy, consistent with the design intent of assigning higher SAM-LoRA weight for crack connectivity.

Effect of class-conditional teacher weighting in DTKD (final model, no SRL). “Equal” uses $w_{T2} = 0.5$ for all classes; “Class-cond.” uses $w_{T2} = 0.6$ for crack, $w_{T2} = 0.3$ for spalling. Bold: better value per metric. All values in %.

Weighting	$mIoU_{fg}$	IoU_{cr}	IoU_{sp}	$BF1_{fg}$	$clDice_{fg}$	$ConnR_{fg}$
Equal (0.5/0.5)	72.2	58.4	86.0	71.5	85.8	83.1
Class-cond.	72.3	58.5	86.2	70.6	86.4	83.0

Teacher selection ablation.

Since the DTKD student does not use SRL, one might expect a teacher trained *without* SRL (higher $ConnR_{fg}$) to provide cleaner knowledge transfer. We test this by replacing Teacher 1 (TopoDistill+SRL, $ConnR_{fg}$ 77.8%) with the SRL-free DSConv-only variant ($ConnR_{fg}$ 80.4%). The resulting student achieves 71.1% $mIoU_{fg}$ **−1.2** below the G1 teacher variant (72.3%). The degradation concentrates on spalling (IoU_{sp} −2.4, $ConnR_{sp}$ −1.4), while crack metrics are largely unaffected. This indicates that SRL-trained Teacher 1 learns richer internal representations—despite its own connectivity regression—that transfer more effectively through soft-label distillation. Teacher quality for KD is not simply a function of the teacher’s own test metrics; the diversity of learned features matters more.

Qualitative results

Fig. [fig:qualitative] presents representative predictions across all three difficulty tiers, comparing SegFormer-B2, TopoDistill, and TopoDistill+DTKD. SegFormer-B2 produces severe over-segmentation (especially on Easy and Hard samples), while TopoDistill substantially improves crack localisation. TopoDistill+DTKD further refines both crack connectivity and spalling coverage, with per-image IoU scores consistently higher than TopoDistill alone.

image

Cross-dataset evaluation

To assess generalisability beyond the training domain, we perform zero-shot transfer evaluation on the S2DS dataset, which contains 743 images (1024×1024) of concrete surfaces with pixel-level annotations for seven defect classes: background, crack, spalling, corrosion, efflorescence, vegetation, and control point. We map S2DS crack and spalling to our label scheme and treat all other classes as background, since they fall outside the DamSegment label space. All images are resized to 512×512 (bilinear for images, nearest-neighbour for masks). The S2DS test split contains 93 images, of which 27 contain only crack, 20 contain only spalling, and 46 are background-only; crack pixels account for 0.12% and spalling for 4.27% of the test area. No fine-tuning is performed: models trained on DamSegment are applied directly.

Cross-dataset evaluation: DamSegment $\rightarrow$ S2DS (zero-shot). $\$^{\backslash}\$$ TopoDistill is the SRL variant (Teacher 1). All models evaluated without TTA. *TTA row shown for reference only. All values in %.

Model	mIoU _{fg}	IoU _{cr}	IoU _{sp}	BF1 _{fg}	clDice _{fg}	ConnR _{fg}
SegFormer-B2	0.1	0.1	0.0	2.8	1.3	49.9
Mask2Former	2.5	0.1	4.8	5.5	21.2	49.7
TopoDistill $\$^{\backslash}\$$	17.1	7.6	26.7	49.6	72.1	69.3
TopoDistill + DTKD	18.7	8.6	28.9	46.6	68.4	70.1
TopoDistill + DTKD + TTA*	18.6	8.4	28.9	45.3	69.0	69.0

Table 6 reveals a large domain gap for all models. SegFormer-B2 and Mask2Former fail almost completely on the unseen domain ($\text{mIoU}_{\text{fg}} < 2.5\%$), indicating severe overfitting to the DamSegment distribution. Models equipped with DSConv achieve substantially higher relative transfer performance ($\text{mIoU}_{\text{fg}} > 17\%$), though the absolute values remain limited, reflecting the inherent difficulty of zero-shot cross-domain segmentation under significant distribution shift. TopoDistill+DTKD achieves the highest mIoU_{fg} (18.7%; TTA does not improve this, yielding 18.6%), while TopoDistill (Teacher 1, with SRL) achieves the highest BF1_{fg} (49.6%) and clDice_{fg} (72.1%).

These results suggest that DSConv’s geometric inductive bias improves cross-domain robustness, as crack and spalling morphology shares common geometric patterns across datasets. DTKD further improves cross-domain transfer (+1.6 mIoU_{fg} over TopoDistill alone), while also maximising in-domain accuracy (+1.9 mIoU_{fg} on DamSegment). Unlike in-domain evaluation, TTA does not improve cross-dataset performance (18.7% $\rightarrow$ 18.6%), suggesting that multi-scale aggregation offers diminishing returns under large domain shift.

Fig. 3 shows representative S2DS predictions. SegFormer-B2 and Mask2Former produce largely incorrect segmentations, while TopoDistill and TopoDistill+DTKD localise defect regions despite the domain shift, consistent with the quantitative results above.

□

Cross-dataset qualitative results on S2DS (zero-shot). Left to right: input, ground truth, SegFormer-B2, Mask2Former, TopoDistill, TopoDistill+DTKD. Red: crack, green: spalling. SegFormer-B2 and Mask2Former fail on the unseen domain, while DSConv-equipped models retain partial defect localisation.

Discussion

DTKD as the critical component.

Beyond the metric gains detailed in Section 4.6, DTKD serves as a powerful training regulariser: the soft-label supervision from two pretrained teachers smooths the loss landscape, reducing cross-seed variance by $\sim 3\text{--}5\times$ (Table 2) and making convergence less sensitive to random initialisation—a practical advantage for deployment scenarios where retraining reliability matters.

Granularity of difficulty-aware modulation.

Inter-teacher disagreement provides a free quality signal that correlates with sample difficulty (+43% KL divergence on Hard vs. Easy). Leveraging this signal at the *pixel level* (confidence-aware KD) is effective, but using it for *sample-level* curriculum learning degrades performance. This suggests that on small datasets, every sample is needed for generalisation, but noisy pixels within a sample can safely be down-weighted—the granularity of the intervention matters more than the quality of the difficulty signal.

SRL: useful for teachers, harmful for students.

SRL forces the teacher to learn richer internal representations that transfer well through distillation, even though SRL degrades the teacher’s own connectivity metrics. As a student training component, however, SRL interferes with DTKD’s soft-label supervision. This asymmetry—that optimal teacher and student training recipes differ—has broader implications for distillation pipeline design.

Indirect use of foundation models.

Rather than directly fine-tuning SAM or DINOv2 (which underperforms task-specific architectures by 2–4% mIoU on small datasets), using them as *teachers* in knowledge distillation yields better results. The student inherits the foundation model’s structural knowledge while maintaining inference efficiency identical to the base model.

Crack–spalling tradeoff.

The persistent tradeoff across all loss-based strategies (Section 4.7) arises from shared geometric boundaries between defect types. Pixel-level loss modification alone cannot resolve the crack precision bottleneck without collateral damage to spalling, suggesting that sampling-based, architectural, or post-processing approaches (e.g., class-specific CRF refinement) are more promising.

Practical implications.

The student model has identical inference cost to the base TopoDistill (96 FPS, 31.3M parameters), with teachers used only during training. The per-tier analysis suggests prioritising deployment on Hard-tier regions where DTKD provides the largest benefit. The model’s connectivity (ConnR_{fg} 83.0%) helps produce predictions directly usable for crack length estimation without extensive post-processing to reconnect fragmented segments.

Failure modes and limitations.

The model underperforms on (i) thin hairline cracks (<2 pixels wide) and (ii) co-occurring crack–spalling regions where it over-predicts one class at the expense of the other. The DamSegment dataset comprises images from a single dam site with patch-level splitting; spatially adjacent patches may share visual characteristics, potentially inflating in-domain metrics. The S2DS cross-dataset transfer partially mitigates this concern but reveals a substantial domain gap (~19% mIoU_{fg}), indicating that zero-shot cross-site transfer remains far from solved. Multi-site datasets and domain adaptation would strengthen generalisability. Per-class teacher weights are set manually; adaptive weighting is future work. Crack IoU remains challenging (~59%) due to extreme class imbalance, and the DSConv branch is crack-specific.

Conclusion

We presented TopoDistill, a topology-aware segmentation framework for concrete dam defects that combines Dynamic Snake Convolution with class-conditional dual-teacher knowledge distillation (DTKD). The two enhancements are complementary: DSConv provides geometric inductive bias for crack connectivity, while DTKD improves pixel-level accuracy and serves as a training regulariser, together achieving state-of-the-art

performance on the DamSegment benchmark (72.3% mIoU_{fg}, 83.0% ConnR_{fg}) at no additional inference cost. An investigation of Skeleton Recall Loss revealed an asymmetric effect in distillation—SRL benefits teachers but harms students—highlighting that optimal training recipes differ across pipeline stages. Exploration of alternative loss functions uncovered a persistent crack-spalling tradeoff at shared class boundaries, and zero-shot cross-dataset evaluation confirmed that DSConv’s geometric prior substantially improves cross-domain robustness (18.7% vs. 0.1% mIoU_{fg}). These findings demonstrate that indirectly leveraging foundation models through knowledge distillation is more effective than direct fine-tuning on small-scale industrial datasets, providing practical guidance for data-scarce inspection scenarios.

Data availability

The DamSegment dataset is publicly available . Source code will be made available upon publication.

Declaration of competing interest

The authors declare no competing interests.

Acknowledgements

[To be added.]

99

Arafin, P., Billah, A.M., Issa, A., 2024. Multi-class concrete defect segmentation using modified DeepLabV3+. Structures 61, 106047.

Benz, C., Rodehorst, V., 2022. Image-based detection of structural defects using hierarchical multi-scale attention. In: DAGM German Conference on Pattern Recognition (DAGM GCPR). Springer. Dataset: <https://github.com/Christoph-Benz/S2DS>.

Chen, L.-C., Zhu, Y., Papandreou, G., Schroff, F., Adam, H., 2018. Encoder-decoder with atrous separable convolution for semantic image segmentation. In: ECCV, LNCS 11211. pp. 801–818.

Chen, B., Yang, M., Zhang, L., 2024. A review of visual perception methods for intelligent dam inspection. Journal of Hydroelectric Engineering 43, 1–15.

Cheng, B., Misra, I., Schwing, A.G., Kirillov, A., Girdhar, R., 2022. Masked-attention mask transformer for universal image segmentation. In: CVPR. pp. 1290–1299.

Clough, J.R., Byrne, N., Oksuz, I., Zimmer, V.A., Schnabel, J.A., King, A.P., 2020. A topological loss function for deep-learning based image segmentation using persistent homology. IEEE Trans. Pattern Anal. Mach. Intell. 44 (12), 8175–8188.

Csurka, G., Larlus, D., Perronnin, F., 2013. What is a good evaluation measure for semantic segmentation? In: BMVC. pp. 1–11.

Dai, J., Qi, H., Xiong, Y., Li, Y., Zhang, G., Hu, H., Wei, Y., 2017. Deformable convolutional networks. In: ICCV. pp. 764–773.

Li, M., Yuan, J., Ren, Q., Luo, Q., Fu, J., Li, Z., 2024. CNN-Transformer hybrid network for concrete dam crack patrol inspection. Automation in Construction 163, 105440.

Zhang, H., Chen, D., Wang, C., 2022. Confidence-aware multi-teacher knowledge distillation. In: ICASSP. pp. 4498–4502.

Furlanello, T., Lipton, Z.C., Tschannen, M., Itti, L., Anandkumar, A., 2018. Born again neural networks. In: ICML. pp. 1607–1616.

- Gao, X., Li, Y., Wang, Z., 2025. CCT Net: A dam surface crack segmentation model based on CNN and Transformer. *Infrastructures* 10 (9), 240.
- Gharehbaghi, V., Bennett, C., Lequesne, R.D., Zhao, H., Li, J., 2026. DamSegment: A curriculum-structured image dataset for damage assessment of concrete dams. *Data in Brief* 66, 112671. <https://doi.org/10.1016/j.dib.2026.112671>
- Hinton, G., Vinyals, O., Dean, J., 2015. Distilling the knowledge in a neural network. In: *NeurIPS Deep Learning Workshop*.
- Hu, X., Li, F., Samaras, D., Chen, C., 2019. Topology-preserving deep image segmentation. In: *NeurIPS*. pp. 5657–5668.
- Hu, E.J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., Chen, W., 2022. LoRA: Low-rank adaptation of large language models. In: *ICLR*.
- Arora, D., Kumar, N., Gupta, S., 2025. Does the skeleton-recall loss really work? *arXiv preprint arXiv:2508.11374*.
- Kirillov, A., Mintun, E., Ravi, N., Mao, H., Rolland, C., Gustafson, L., Xiao, T., Whitehead, S., Berg, A.C., Lo, W.-Y., Dollár, P., Girshick, R., 2023. Segment anything. In: *ICCV*. pp. 4015–4026.
- Liu, Y., Yao, J., Lu, X., Xie, R., Li, L., 2019. DeepCrack: A deep hierarchical feature learning architecture for crack segmentation. *Neurocomputing* 338, 180–191.
- Liu, Y., Chen, K., Liu, C., Qin, Z., Luo, Z., Wang, J., 2019. Structured knowledge distillation for semantic segmentation. In: *CVPR*. pp. 2604–2613.
- Liu, H., Miao, X., Mertz, C., Xu, C., Kong, H., 2021. CrackFormer: Transformer network for fine-grained crack detection. In: *ICCV*. pp. 3783–3792.
- Pan, Z., Zhang, X., Jiang, Y., Li, N., Mei, C., Su, H., Cai, Y., 2025. Automated detection system of metro tunnel lining crack using dynamic snake convolution. *Comput.-Aided Civ. Infrastruct. Eng.* 40, e70065.
- Joo, S., Kim, H., 2026. An implementation of the crack topology score with extensions. *arXiv preprint arXiv:2601.10762*.
- Oquab, M., Darcet, T., Moutakanni, T., Vo, H., Szafraniec, M., Khalidov, V., Fernandez, P., Haziza, D., Massa, F., El-Nouby, A., Howes, R., Huang, P.-Y., Xu, H., Sharma, V., Li, S.-W., Galuba, W., Rabbat, M., Assran, M., Ballas, N., Synnaeve, G., Misra, I., Jegou, H., Mairal, J., Labatut, P., Joulin, A., Bojanowski, P., 2024. DINOv2: Learning robust visual features without supervision. *Trans. Mach. Learn. Res. (TMLR)*.
- Pantoja-Rosero, B.G., Oner, D., Koziński, M., Achanta, R., Fua, P., Pérez-Cruz, F., Beyer, K., 2022. TOPO-Loss for continuity-preserving crack detection using deep learning. *Construction and Building Materials* 344, 128264.
- Qi, Y., He, Y., Qi, X., Zhang, Y., Yang, G., 2023. Dynamic snake convolution based on topological geometric constraints for tubular structure segmentation. In: *ICCV*. pp. 6070–6079.
- Salehi, S.S.M., Erdogmus, D., Gholipour, A., 2017. Tversky loss function for image segmentation using 3D fully convolutional deep networks. In: *MLMI Workshop, MICCAI*. Springer, LNCS 10541. pp. 379–387.
- Ronneberger, O., Fischer, P., Brox, T., 2015. U-Net: Convolutional networks for biomedical image segmentation. In: *MICCAI*, LNCS 9351. pp. 234–241.
- Shit, S., Paetzold, J.C., Sekuboyina, A., Ezhov, I., Unber, A., Zhylka, A., Pluim, J.P.W., Bauer, U., Menze, B.H., 2021. cIDice – A novel topology-preserving loss function for tubular structure segmentation. In: *CVPR*. pp. 16560–16569.
- Shit, S., Koner, R., Paetzold, J.C., Ezhov, I., Menze, B.H., 2024. Skeleton recall loss for connectivity conserving and resource efficient segmentation of thin tubular structures. In: *ECCV*, LNCS 15118. pp. 363–380.

- Liang, P., Huang, H., Pu, B., Chen, J., Hua, X., Zhang, J., Ma, W., Chen, Z., Li, Y., Chang, Q., 2025. Task-specific knowledge distillation from the vision foundation model for enhanced medical image segmentation. arXiv preprint arXiv:2503.06976.
- Wang, Y., Zhou, W., Jiang, T., Bai, X., Xu, Y., 2020. Intra-class feature variation distillation for semantic segmentation. In: ECCV, LNCS 12370. pp. 346–362.
- Xie, E., Wang, W., Yu, Z., Anandkumar, A., Alvarez, J.M., Luo, P., 2021. SegFormer: Simple and efficient design for semantic segmentation with transformers. In: NeurIPS. pp. 12077–12090.
- Yu, K., Chen, I.-M., Wu, J., 2024. DSCformer: A dual-branch network integrating enhanced dynamic snake convolution and SegFormer for crack segmentation. arXiv preprint arXiv:2411.09371.
- You, S., Xu, C., Xu, C., Tao, D., 2017. Learning from multiple teacher networks. In: KDD. pp. 1285–1294.
- Yuan, L., Tay, F.E., Li, G., Wang, T., Feng, J., 2021. Revisiting knowledge distillation via label smoothing regularization. In: CVPR. pp. 3903–3911.
- Yé, Z., Lovell, L., Faramarzi, A., Ninic, J., 2024. Sam-based instance segmentation models for the automation of structural damage detection. *Advanced Engineering Informatics* 62, 102826.
- Zou, Q., Zhang, Z., Li, Q., Qi, X., Wang, Q., Wang, S., 2019. DeepCrack: Learning hierarchical convolutional features for crack detection. *IEEE Trans. Image Process.* 28 (3), 1498–1512.